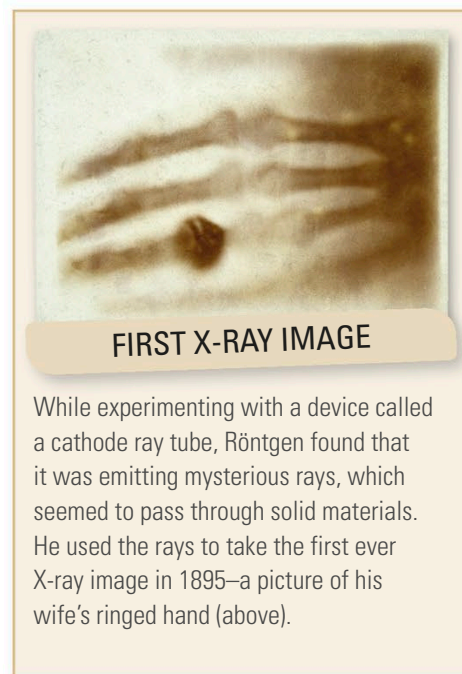


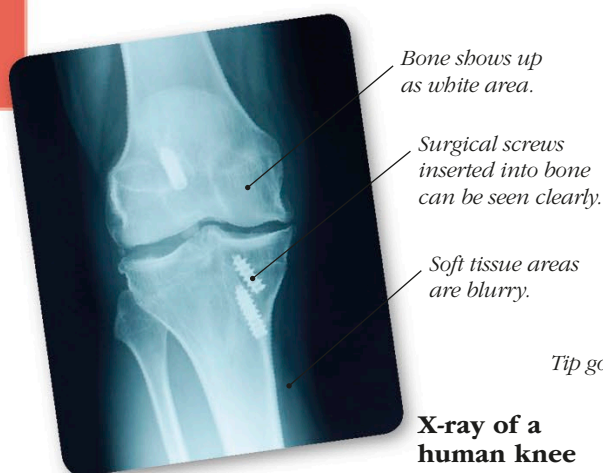
Looking inside

Before 1895, looking inside a patient's body meant cutting it open. The discovery of X-rays by the German scientist Wilhelm Röntgen provided a new way. X-rays are a form of electromagnetic radiation (like light) that can pass through softer parts of the body, such as organs, but are absorbed by denser parts, such as bones, which then show up clearly on X-ray images. Many other methods have since been invented to look inside the body safely.



FIRST X-RAY IMAGE

While experimenting with a device called a cathode ray tube, Röntgen found that it was emitting mysterious rays, which seemed to pass through solid materials. He used the rays to take the first ever X-ray image in 1895—a picture of his wife's ringed hand (above).



Bone shows up as white area.

Surgical screws inserted into bone can be seen clearly.

Soft tissue areas are blurry.

X-ray of a human knee

NAMING THE UNKNOWN

Röntgen didn't know what the radiation was, so he called it "X-ray" ("X" stands for "unknown"). His discovery won him a Nobel Prize in Physics in 1901. Today, X-rays are used by doctors to check for broken bones or foreign objects, including surgical implants, inside the body.



SHINING A LIGHT

The American doctors Basil Hirschowitz and Larry Curtiss invented the fiber-optic endoscope in 1957. It is a thin, flexible tube that is filled with glass or plastic fibers along which light signals can be transmitted. The device is inserted into the patient's body to relay images of its insides to the doctors.



Tip goes inside the body.

Markings show how far the endoscope has gone inside the body.

A light source is attached here.

Dial moves the tip of the endoscope to change the view.

Operator looks through the eyepiece.

Endoscope

IMAGES FROM SOUND

First used in the 1950s, ultrasound scanners send high-pitched sounds into the body. Different tissues, such as bone or muscle, produce different echoes, creating a two-dimensional "sound picture," known as a sonogram. This process is harmless and is often used for scanning babies in the womb (left).